Elijah Croft

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Education

California State University Fullerton — Fullerton, CA

Double Major: B.S. Computer Science, B.A. Philosophy — *Anticipated Graduation: May 2028*

- Research: AI Ethics (forthcoming publication); Presented panel on algorithmic decision-making.

Experience

Software Engineer (Contract) — May. 2025 – Present

2E4ME — *Newport Beach, CA*

- Designed and deployed a full-stack scheduling platform serving 20+ instructors and 70+ students, streamlining timetable management and reducing manual scheduling effort.
- Implemented a constraint-based optimization engine using OptaPlanner to automatically generate timetables.
- Designed subdomain infrastructure with Google OAuth 2.0 SSO, enabling secure internal authentication.
- Designed and delivered robotics and programming lessons covering programming and robotics.

Robotics Coach/Programming Instructor — Mar. 2025 - Present

Magikid Robotics Lab — *Tustin, CA*

- Led a team of 10 to the VEX AI World Championships, achieving 18th place among 40 international teams.
- Designed and deployed 3-tier skills assessment system covering mechanical engineering, sensor integration, and algorithmic thinking, used to evaluate 50+ students across 12-week program cycles
- Assisted with AI integration in youth robotics education by implementing autonomous navigation based on the Markov Decision Process, reducing path planning errors by 80% in competition scenarios

Projects

MixSafe | *Next.js, TypeScript, Tailwind CSS, Django* | **Demo** | **GitHub** — Oct. 2024 - Present

- Built and deployed a harm-reduction platform serving 500+ weekly users and processing 2,000+ monthly interaction checks across 15+ substance categories.
- Developed a structured interaction database enabling fast, real-time lookups for substance combinations and effects.
- Led a multi-person development team responsible for platform architecture, data curation, and feature development.

PPP Simulator (Best Use of ArcGIS) | *Next.js, TypeScript, ArcGIS SDK, Leaflet* | **Demo** | **GitHub** — Apr. 2026

- Developed a browser-based ocean pollution prediction platform that models the movement of plastic, oil, and chemical contaminants using current-field vector simulation and geospatial visualization.
- Engineered the core drift-simulation engine by processing real ArcGIS ocean-current data, applying vector-field calculations to predict pollutant trajectories and visualize long-term environmental impact.
- Integrated ArcGIS Maps SDK and Leaflet to render interactive marine hazard overlays/ocean-current visualizations.

C++ Chess Engine | *C++, Makefile, Git, Bitboards* — Jan. 2026 – Present

- Designed and developed a high-performance chess engine in C++ using bitboards and bitwise operations for efficient board representation and move generation.
- Implemented alpha-beta pruning, move ordering, and transposition tables to optimize evaluation and search.
- Developed adjustable engine-strength controls and complete legal move generation, including castling and en passant.

Automated Delivery Robot | *C++, Python, ROS2, Raspberry Pi 5, LiDAR, CAD* — Aug. 2025

- Engineered an autonomous ROS2-based robot featuring LiDAR localization, navigation, and obstacle avoidance.
- Designed power system with a 12V rechargeable ion battery, BMS, and motor driver; built aluminum chassis

ScuffedAI Discord Bot | *Python, OpenAI API, Discord API, Linux* — Mar. 2023

- Developed a personality-driven Discord chatbot using the OpenAI API and custom agent configurations to emulate streamer xQc, generating 1,000+ context-aware responses across 5 servers with 100+ active users.
- Implemented persistent conversational context and deployed the application on a self-hosted Arch Linux server.

Technical Skills

Languages: Python, C++, JavaScript/TypeScript, Java, Scala
Tools: Git, Docker, Linux, Bash
Embedded: ROS2, Raspberry Pi, Jetson Orin Nano, LiDAR, Microcontrollers